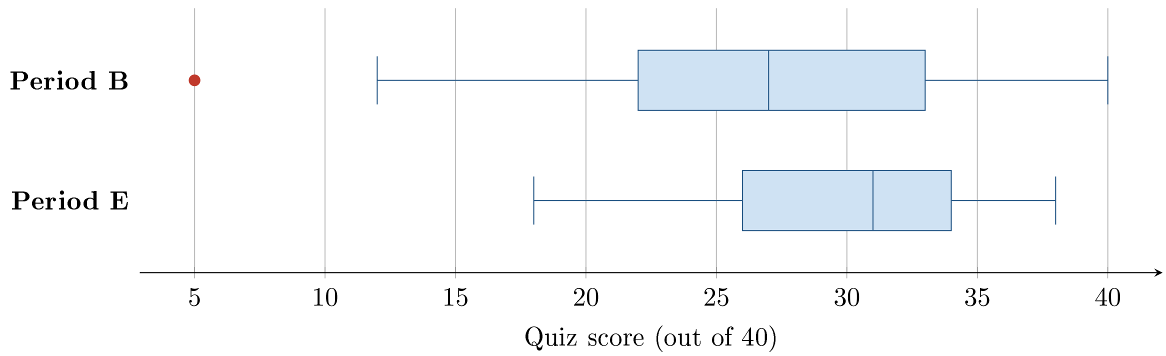
NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

Which Class Did Better?

AP Statistics · Unit 1 · Topic 1.9 · B: 2026-09-24 · E: 2026-09-25

[STAR] TODAY'S PROBLEM

Two AP Statistics classes took the same 40-point quiz. The side-by-side boxplots show the scores. Two students read them differently. **Sam:** "Period B did better — its highest score is 40 and Period E's is only 38." **Tia:** "Period E did better — its median is higher and its scores are more consistent."



FIRST TAKE — before we discuss (5 min)

Which class did better on the quiz? One sentence, and name ONE feature of the plot you used.

[BOOK] COMPARING DISTRIBUTIONS (keep on the page)

Use comparative words such as higher, lower, more variable, or similar. Compare center (median), variability (IQR $= Q_3 - Q_1$, range = maximum minus minimum), apparent shape, and unusual features in context. Include plotted outliers when finding the overall range. A boxplot hides values within each quartile, so qualify shape claims with 'appears.' A median describes center; a maximum describes the highest observed value.

Work (a)–(c) from the rules box:

DOK 1 **(a)** Read the median for each class.

DOK 2 **(b)** In two or three sentences, compare the quiz scores' center, spread, apparent shape, and unusual features. Use points and comparative words.

DOK 3 **(c)** Whose claim is better supported, Sam's or Tia's? Cite two specific features of the boxplots that settle it. Then: if you could report only ONE statistic per class to a parent asking 'how did the class do?', which statistic would you choose, and why that one rather than the maximum?

Frame: ________'s claim is better supported because the ________ of Period E is ________ than Period B's, and Period E's scores are ________.

Frame: I would report the ________ because the maximum describes ________________.

EXIT — turn this in

One thing the rules box changed about my first take: ________________________________

Optional #1 — not collected

DOK 2 Two bus routes were timed for 20 mornings each. Route 1: mean 24.0 min, SD 6.5 min, median 22 min. Route 2: mean 24.5 min, SD 1.8 min, median 24 min. Write two sentences comparing the routes for a student who must not be late, using comparative language and the statistics given.